

Façade Element Detection for Continuous Value Engineering

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1 · Executive summary

Group 02's Final Master's Project is an **AI-Powered BIM Decision Support System for Continuous Value Engineering** — it reads the BIM model to tell a design team where cost is accumulating while the design is still moving. Three situations put that truth outside the model: at concept stage no façade has been modelled yet; a precedent or competitor building has no model at all; and an as-built façade may have diverged from the model on site.

This unit supplies the **perception component**. A YOLOv8n detector trained on a five-class façade contract returns the envelope elements present in a photograph or rendered elevation, from which the VE engine derives Window-to-Wall Ratio and relative envelope cost — for a building it has no model of. The envelope is the right target because it is typically **15–25 % of construction cost** and the first package opened in a VE exercise.

Success criteria. Recover Window-to-Wall Ratio to within **±10 %** of a human's reading of the same image, on frontal-to-30°-oblique daylight elevations of 3–10 storey buildings, in under **2 seconds** per image.

The one non-obvious decision is the headline metric. Because the VE read-out is computed from **box area**, a box covering the right window but 20 % too large scores a clean hit at IoU 0.50 while producing a 20 % cost error — **mAP50** is blind to precisely the failure that costs this project money. The headline metric is therefore **mAP50-95**, and localization drift is reported as a separate error category rather than folded into the true positives.

2 · Method

Class contract

Five classes with positional ids, enforced at load time: `window`, `curtain_wall`, `door`, `balcony`, `louvre_screen`. The `window / curtain_wall` boundary is resolved by one procedural test — *can I trace unbroken glass across it?* — chosen so two annotators cannot disagree. Three standing rules: no reflections; no neighbouring buildings; partial and occluded elements are labelled to an 8 px floor. Boxes run to the outer frame edge, because the frame is part of the purchased unit.

Pipeline

Transfer learning from COCO with `yolo8n.pt`; **30 epochs, batch 16, imgsz 640, seed 42**, early stopping at patience 20. Split **80 / 20** in Roboflow, so it is recorded against an immutable dataset version. Every parameter lives in `src/config.py` and is read from there by the notebooks, so the documentation cannot drift from the run; metrics files are generated by `src/train.py`, and nothing in the repository quotes a number that was typed.

SAM exploration

SAM accelerated `curtain_wall`, `door` and `balcony` — large convex regions with strong edges — by roughly 25–30 % of annotation time, conditional on human review of every mask. It failed on the `window / curtain_wall` boundary, because that distinction is a semantic rule about buildings, not a visual edge. More seriously, it segments mirror-glass spandrel panels beautifully — those are opaque and must not be labelled at all, so unreviewed auto-labelling would have injected this project's most expensive error class at scale, with tight boxes and the appearance of quality. The useful outcome was the contract, not the time saved: the unbroken-glass test exists *because* SAM exposed the ambiguity.

3 · Results

0.975		0.946		0.948		0.879	
PRECISION		RECALL		MAP50		MAP50-95 · HEADLINE	
CLASS	VAL INST.	PRECISION	RECALL	MAP50	MAP50-95		
window	316	0.980	0.951	0.985	0.951		
curtain_wall	37	0.980	1.000	0.995	0.976		
door	24	0.952	0.830	0.821	0.809		
balcony	66	0.994	0.970	0.965	0.897		
louvre_screen	48	0.971	0.979	0.971	0.764		

Verification run — not a performance claim. These figures come from the procedural dataset in `src/synthetic_facades.py`, which exists so the notebooks run end to end for a reader with no Roboflow account, no API key and no GPU. Synthetic façades are axis-aligned and synthetically lit; the domain gap to a photograph is large. The banner is generated from `DATASET_SOURCE`, so it cannot be left on by accident or removed by hand.

4 · Interpretation — three findings

The metric you report decides what you think you built

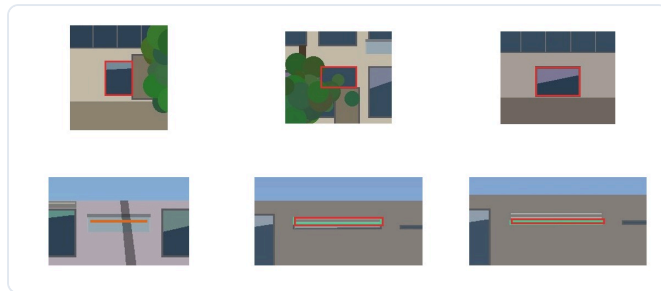
`louvre_screen` scores mAP50 **0.971** and mAP50-95 **0.764** — a 21-point collapse. The model finds almost every louvre screen and boxes almost none tightly; worst observed case **+31 % box area**. Thin, high-aspect-ratio elements suffer roughly constant *absolute* regression error, which on a 12 px-tall element is 25 % of the dimension. Every one of those boxes is a true positive at IoU 0.50, so a report quoting mAP50 would record this as the second-best class. Measured strictly it is the worst — and in a read-out computed from area, a +31 % box is a +31 % cost.

Class imbalance appears as a systematic cost bias, not as noise

At **13.8 : 1** between most and least frequent class, `door` — 96 training instances against 1 341 windows — is worst on every measure (recall 0.830, mAP50 0.821). Both `door` → `window` confusions had near-perfect boxes (IoU 0.976, 0.970): localised right, *named* wrong — a classifier problem caused by imbalance, not a localisation one. Since `door` carries the second-highest unit rate, missing one in six understates entrance cost systematically, so it does not average out across a portfolio.

The dangerous error is the quiet one

False negatives outnumber false positives 23 to 9. A false positive is challenged the moment a human looks at the annotated image; a false negative produces a plausible number nobody questions. That inverts the PPE logic, where recall dominates because a miss is a safety event. Here both directions distort a *ratio*, and the miss is invisible — which is why every evidence render shows the boxes, not only the numbers. The human check is the mitigation.



Worst false positives, false negatives, class confusions and localization drifts, ranked automatically by `src/error_analysis.py` over the whole validation split. Green = ground truth, red = wrong prediction, orange = missed instance.

ERROR TYPE	COUNT	% OF ERRORS
False negative	23	56.1 %
False positive	9	22.0 %
Localization error	6	14.6 %
Class confusion	3	7.3 %

Iteration plan — data, not a bigger model. Every failure above is a data failure; a larger model trained on the same gaps learns the same wrong cues with more capacity.

- 150 hard negatives** of reflective opaque surfaces, added unlabelled. The model learned "reflection gradient ⇒ glazing" and has never seen a counterexample — the only fix thresholding cannot substitute for, since the model is confidently wrong, not uncertain. *Measure*: zero-IoU `window` false positives fall from 2 to ≤ 1 per 24 images, with no loss of recall.
- Rebalance `door` 96 → ~300 instances** by targeted collection, not oversampling — oversampling amplifies whatever is unrepresentative about the existing 96. *Measure*: recall ≥ 0.92, imbalance below 6 : 1.
- Small-instance coverage plus an evidenced resolution decision.** Sixteen `window` misses cluster below 600 px². Add ~100 small-instance images, then compare 640 against 1280 and tiled inference — doubling resolution quadruples training time, so this is a decision to evidence, not assume.

5 · Limitations

This model is an assistive tool for preliminary screening only. It produces False Negatives. It must NOT be used as the sole verifier for life-safety decisions.

- Not a quantity take-off.** Outputs are uncalibrated pixel areas; without a known scale they are not square metres and cannot become square metres — an unsolved problem, not an engineering detail. Every output carries this basis string in its JSON.
- Not a compliance check, and not a BIM-vs-reality verifier** — it cannot measure a ratio in real units, and implements no camera pose or registration.
- Unmeasured outside its training distribution** — heritage and vernacular typologies, night, fog, extreme obliquity, heavy scaffolding. Unmeasured ≠ adequate.
- Synthetic-to-real gap.** Published metrics describe the pipeline, not façade detection. Expect materially lower figures on photographs.
- The validation set does double duty** as early-stopping signal and reported metric, so figures are mildly optimistic. The unseen-image pack is the mitigation.

6 · Governance & licensing

Privacy by design. There is no `person` class and no `vehicle` class. Data minimisation is applied *at the class contract*, so the model is structurally incapable of identifying anyone — not filtered afterwards. No biometric processing, and no client or employer imagery at any point.

Human in the loop. Detections are a first pass reviewed by a qualified architect or cost consultant before any figure reaches a client or a cost plan; source image, weights version, threshold, timestamp and reviewer are logged. The model screens; it certifies nothing. **Licence:** Apache-2.0, chosen over MIT for the explicit patent grant, with dataset rights public / licensed under a per-source table. **Upstream caveat:** Ultralytics YOLOv8 is AGPL-3.0 and fine-tuned weights inherit it, so commercial deployment requires AGPL compliance or an Enterprise Licence — the most commonly missed licensing fact in YOLO projects, recorded in `NOTICE`.